

- ▶ Introduction
- ▶ VPC Network Access Analyzer
- ▼ **VPC Reachability Analyzer**
 - Lab 1 - Troubleshoot the connectivity issue between App server and Database Server
 - Lab 2 - Troubleshoot the connectivity issue between Prod EC2 instance in Prod VPC and Public IP address
- ▶ AWS Network Manager
- ▶ Summary and Resources

Workshop catalog in AWS Builder Center [↗](#)

VPC Reachability Analyzer

VPC Reachability Analyzer

In this workshop you will learn how Amazon VPC Reachability Analyzer helps you to perform connectivity testing between a source resource and a destination resource in your virtual private clouds (VPCs) and how you can troubleshoot connectivity issues caused by netwrok misconfiguration.

Previous

Next

Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose “Customize” or “Decline” to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose “Accept” or “Decline.” To make more detailed choices, choose “Customize.”

Accept

Decline

Customize

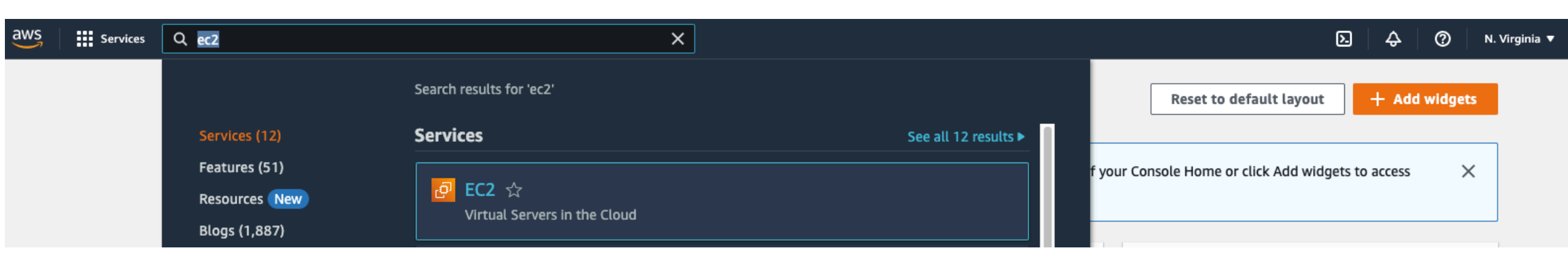
Lab 1 - Troubleshoot the connectivity issue between App server and Database Server

Objective: In this section of the workshop we will use Amazon VPC Reachability Analyzer to troubleshoot and fix connectivity issue between your application and database.

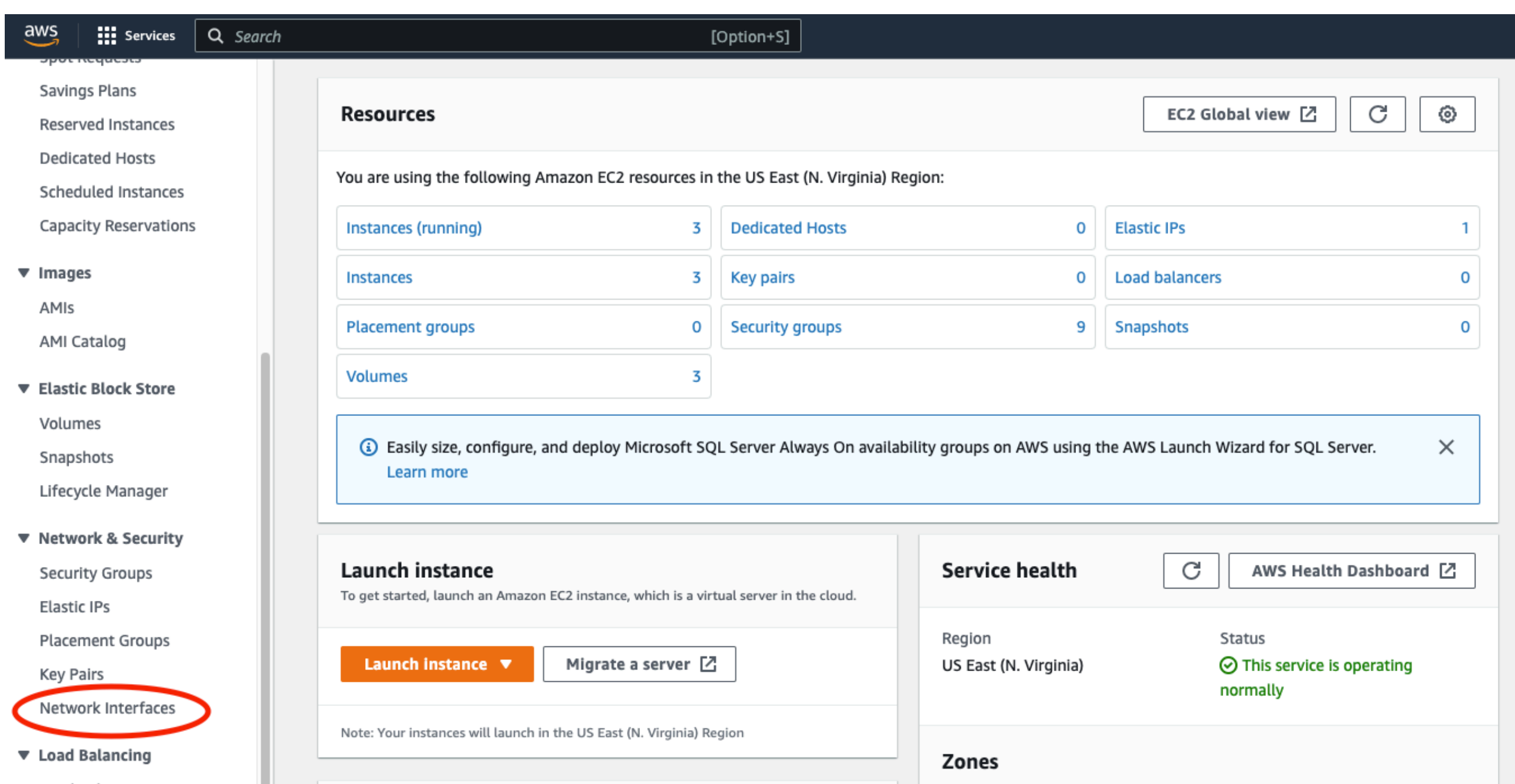
Prerequisite

Find out the **Network interface id** for the database server.

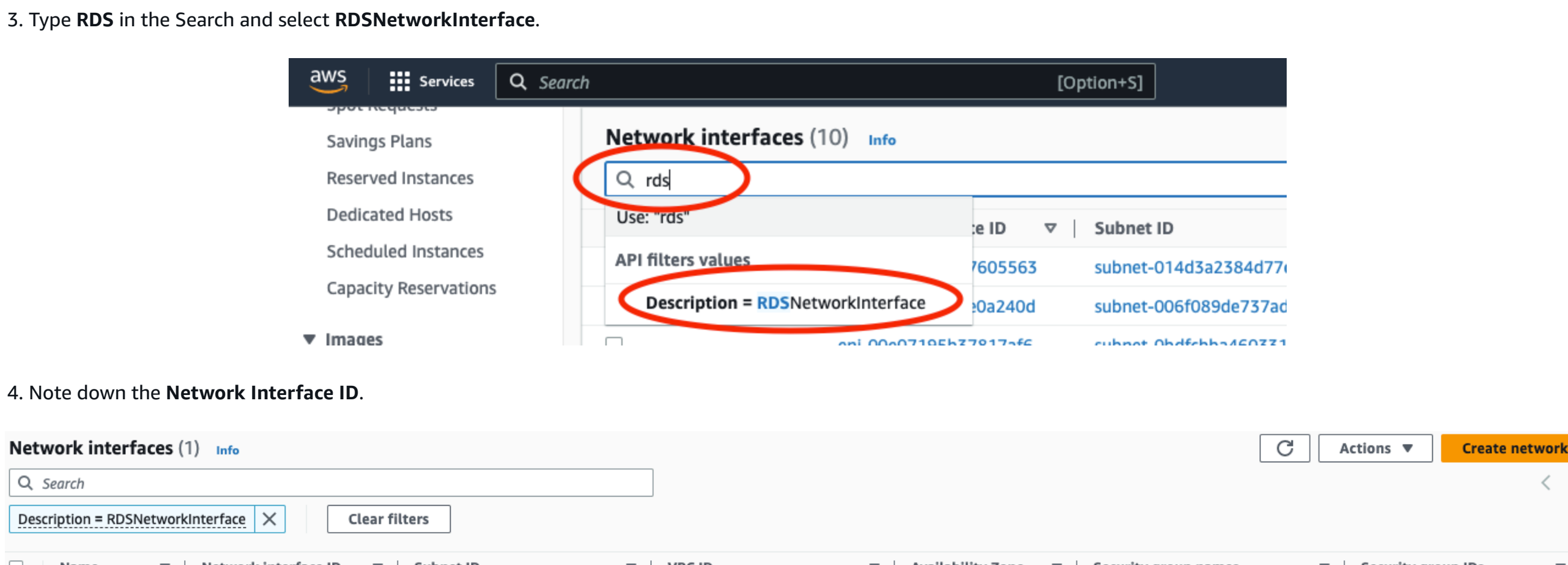
- From the **AWS Console** click in the search bar in the top left corner. Type **EC2** in the services search bar. Select **EC2** from the list.



- Click **Network interface** on the left side of the screen which can be found under the **Network and Security** section.

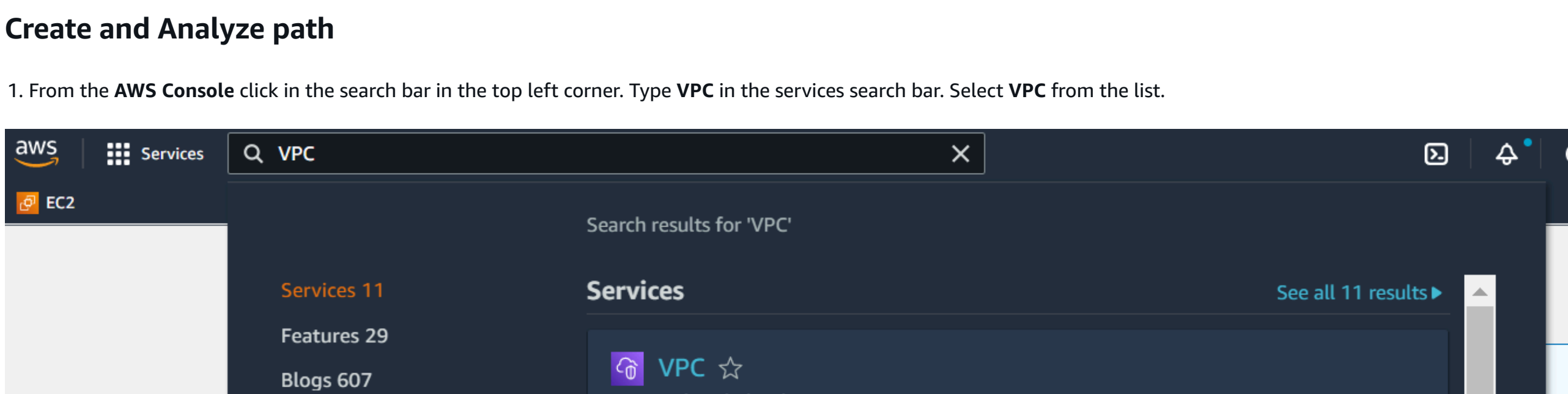


- Type **RDS** in the Search and select **RDSNetworkInterface**.

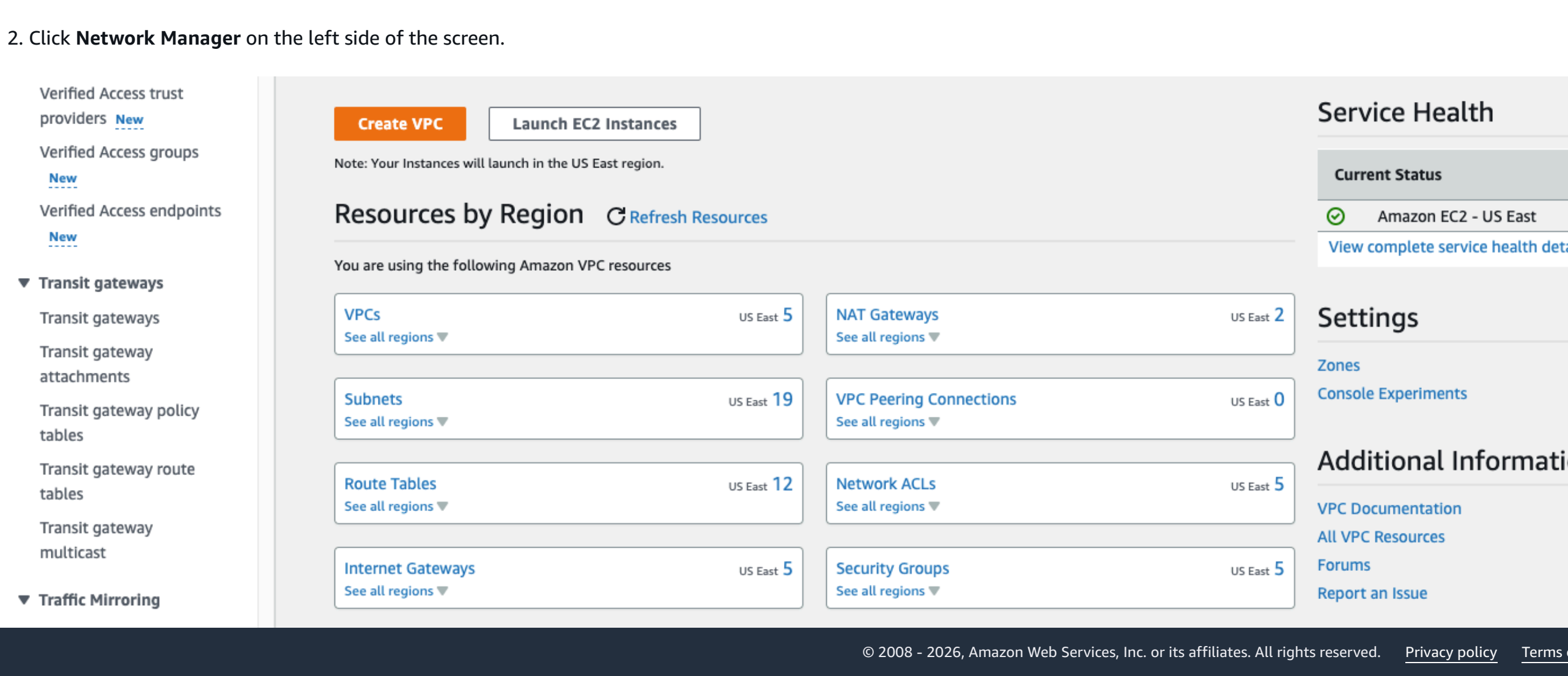


Create and Analyze path

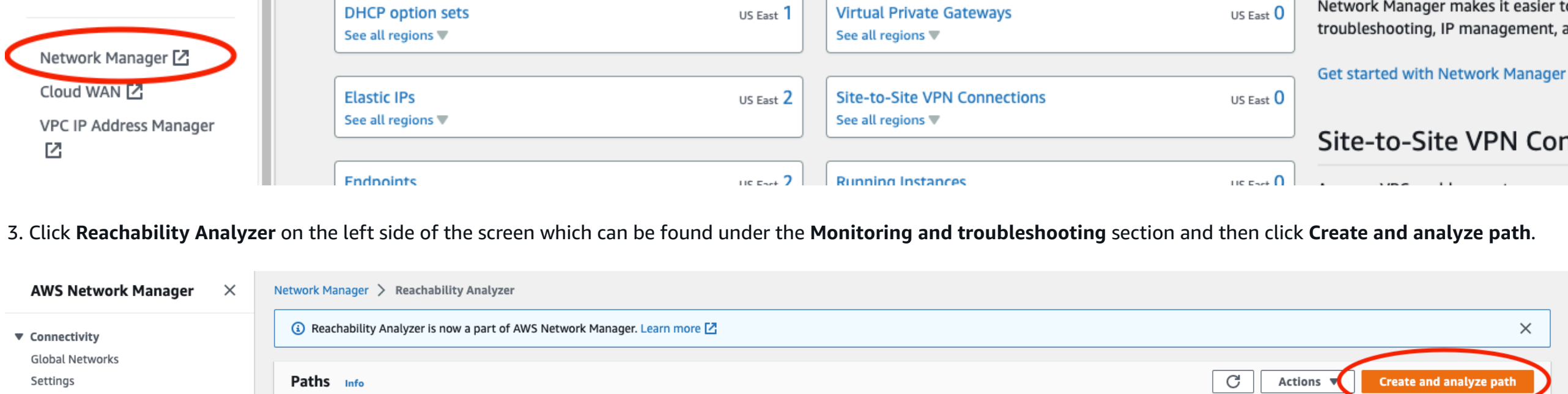
- From the **AWS Console** click in the search bar in the top left corner. Type **VPC** in the services search bar. Select **VPC** from the list.



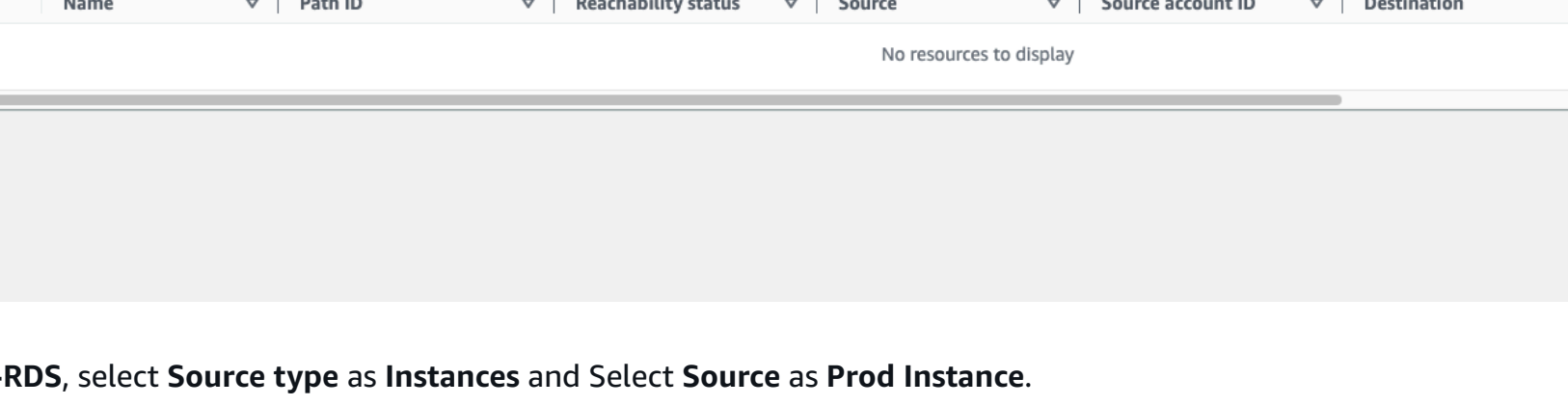
- Click **Network Manager** on the left side of the screen.



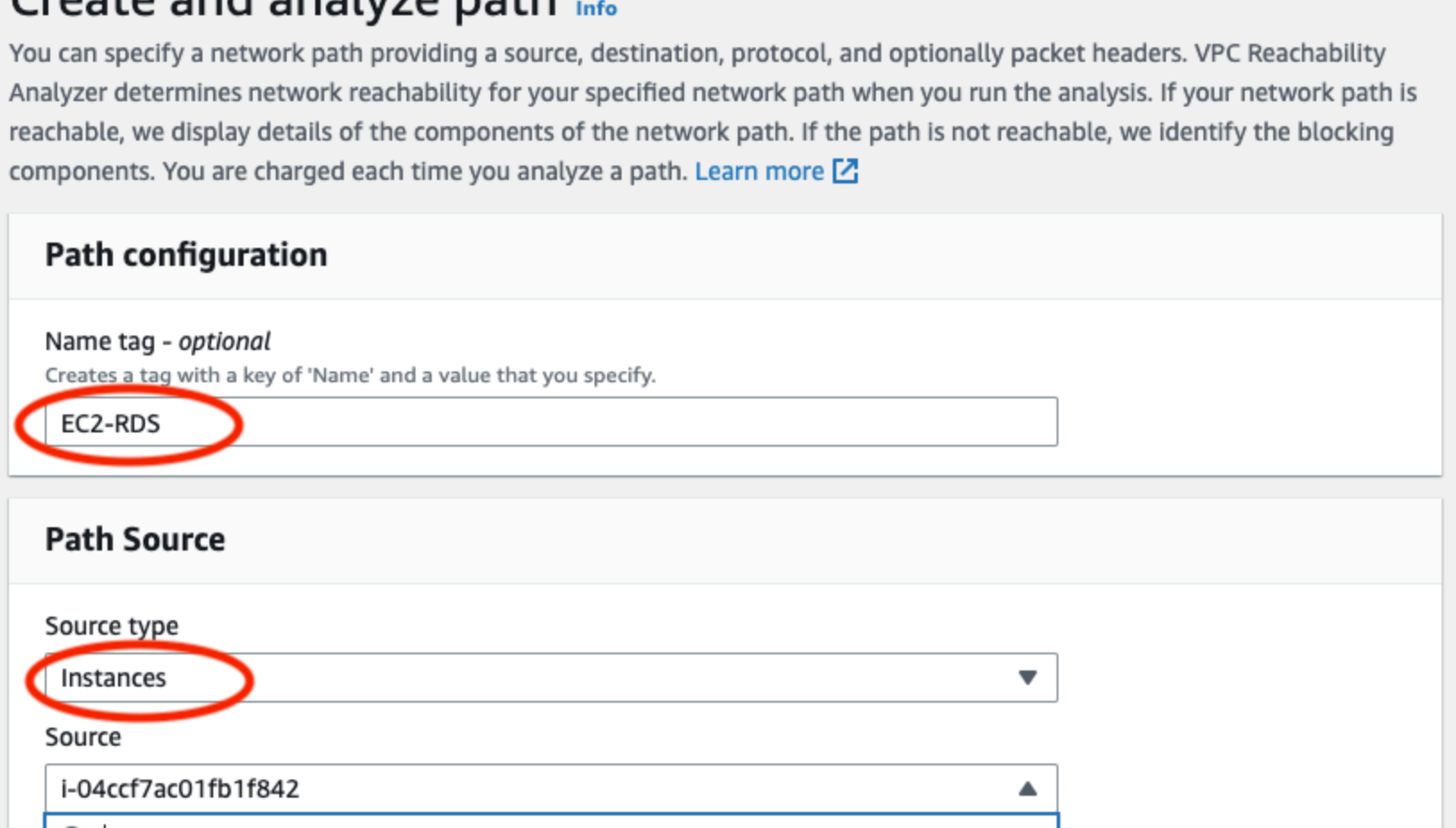
- Click **Reachability Analyzer** on the left side of the screen which can be found under the **Monitoring and troubleshooting** section and then click **Create and analyze path**.



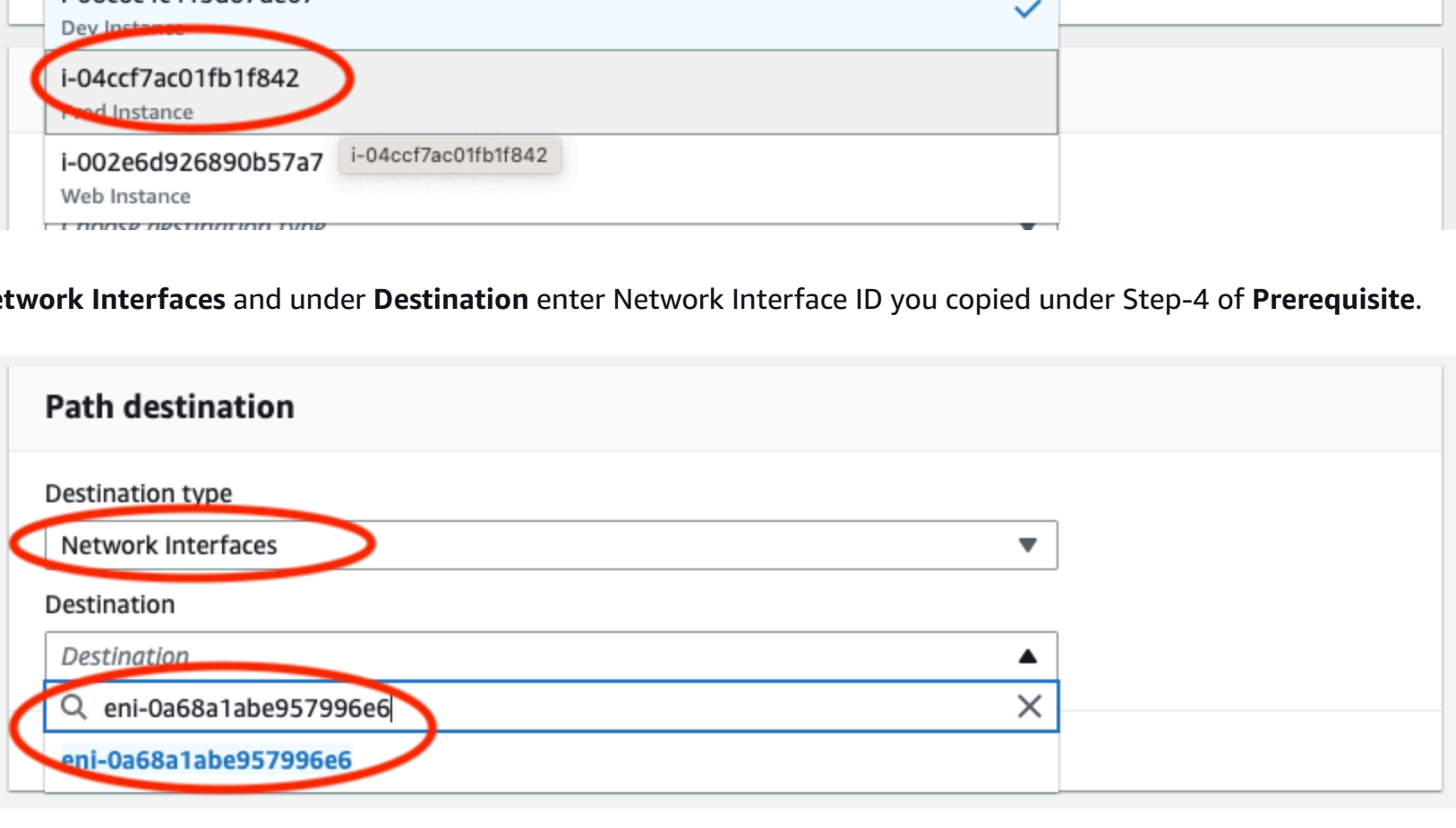
- Enter **Name tag** value as **EC2-RDS**, select **Source type** as **Instances** and Select **Source** as **Prod Instance**.



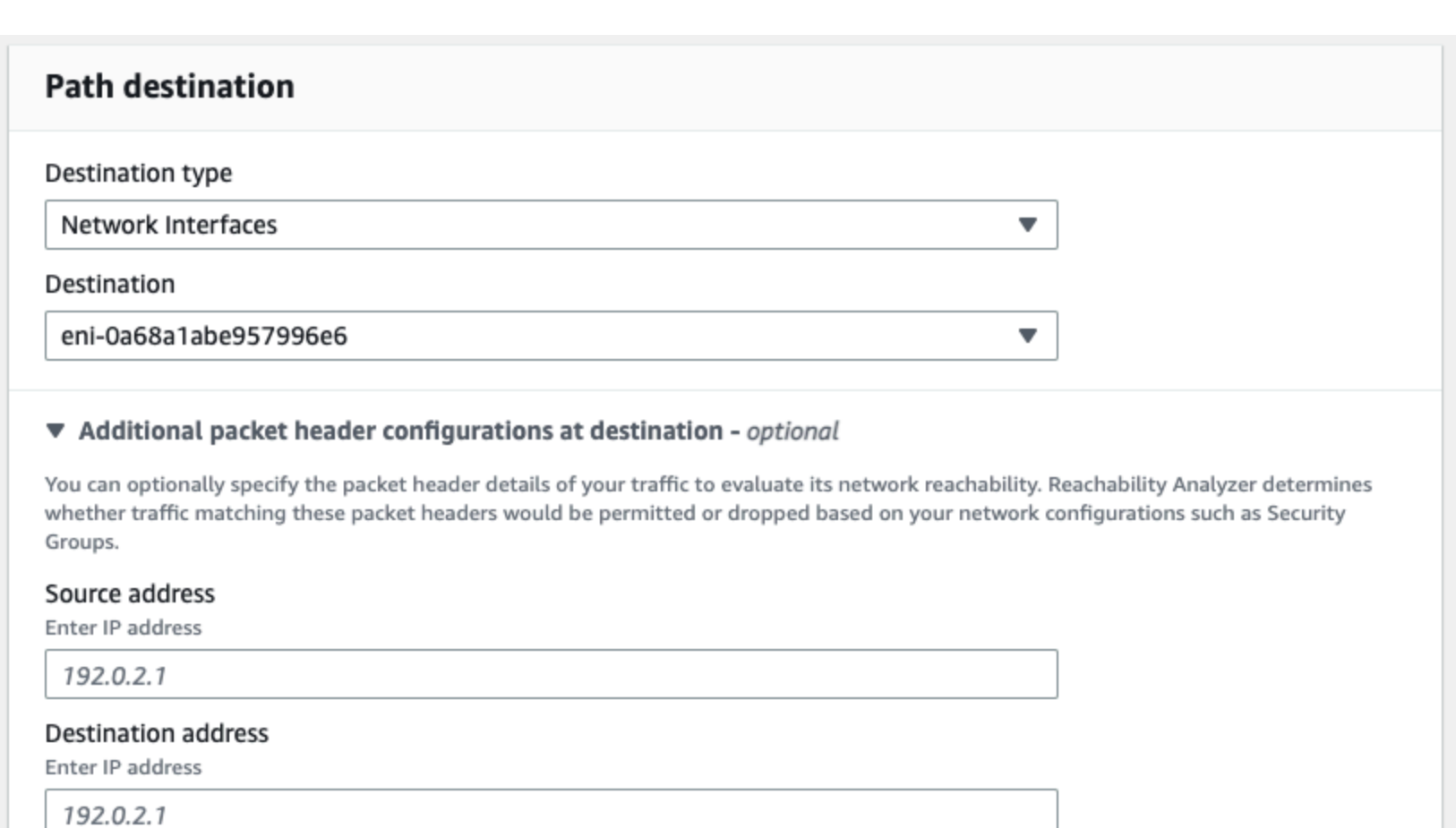
- Select **Destination type** as **Network Interfaces** and under **Destination** enter Network Interface ID you copied under Step-4 of **Prerequisite**.



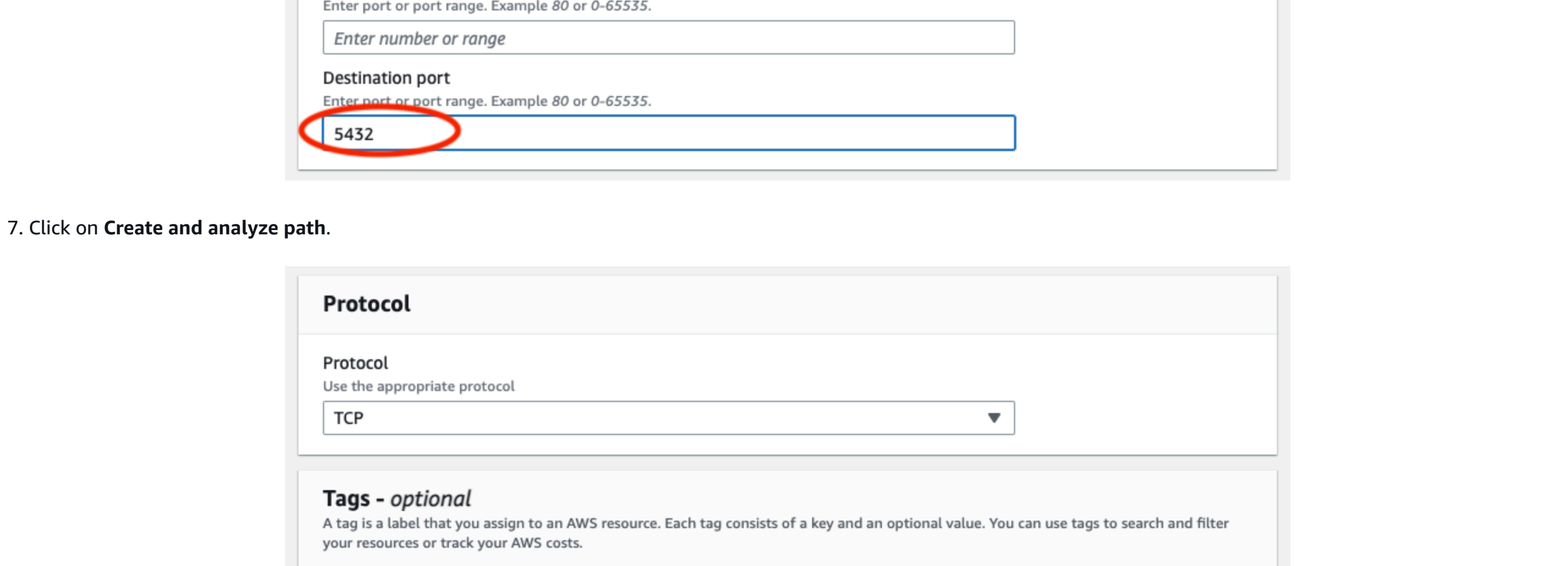
- Enter **Destination port** as **5432**.



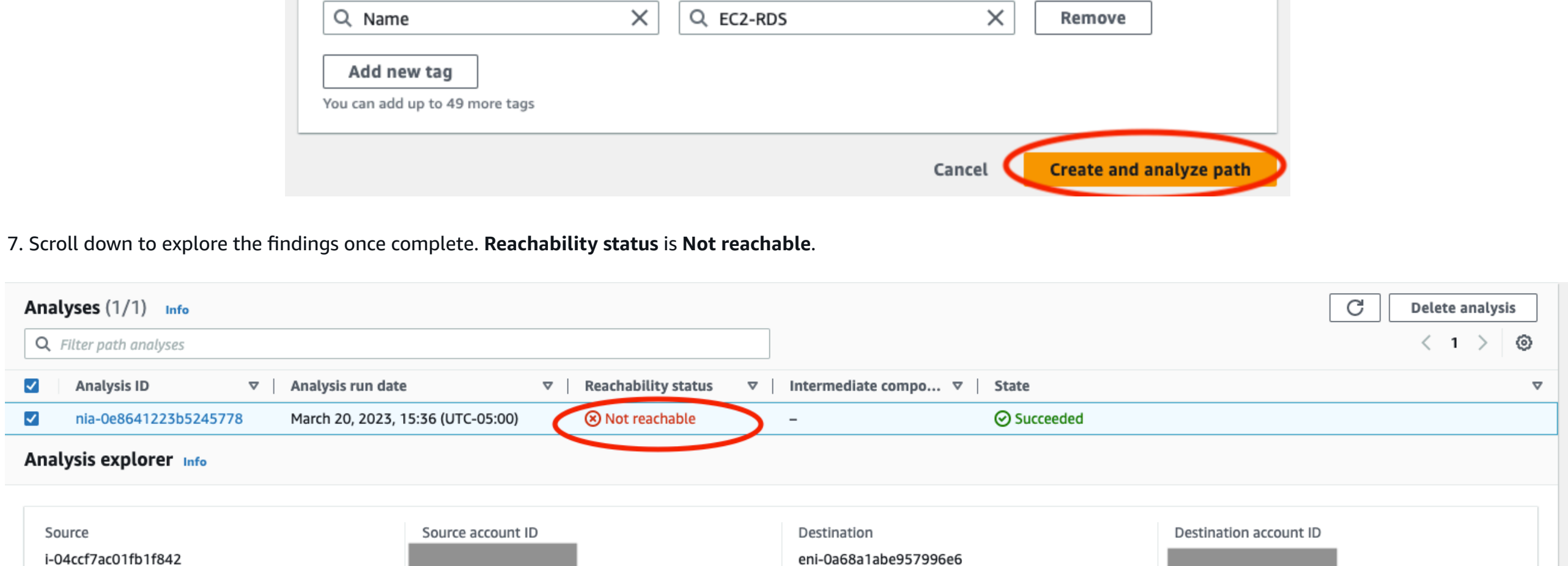
- Click on **Create and analyze path**.



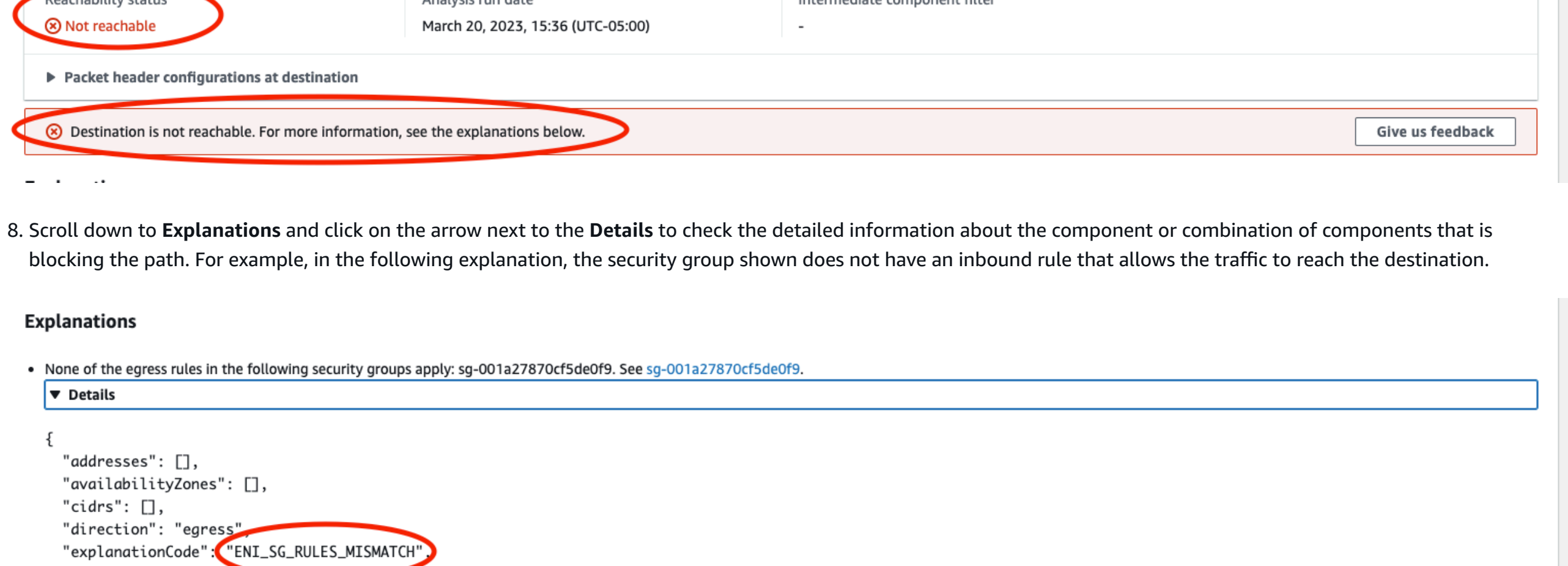
- Scroll down to explore the findings once complete. **Reachability status** is **Not reachable**.



- Scroll down to **Explanations** and click on the arrow next to the **Details** to check the detailed information about the component or combination of components that is blocking the path. For example, in the following explanation, the security group shown does not have an inbound rule that allows the traffic to reach the destination.

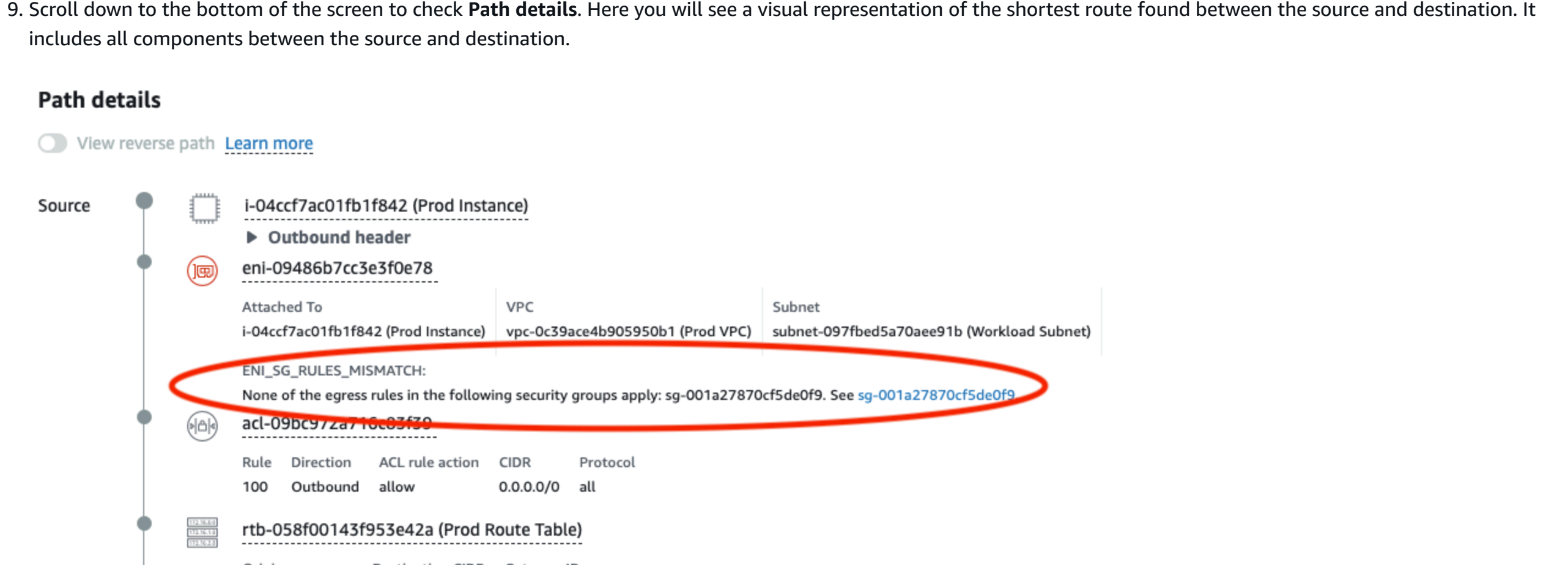


- Scroll down to the bottom of the screen to check **Path details**. Here you will see a visual representation of the shortest route found between the source and destination. It includes all components between the source and destination.

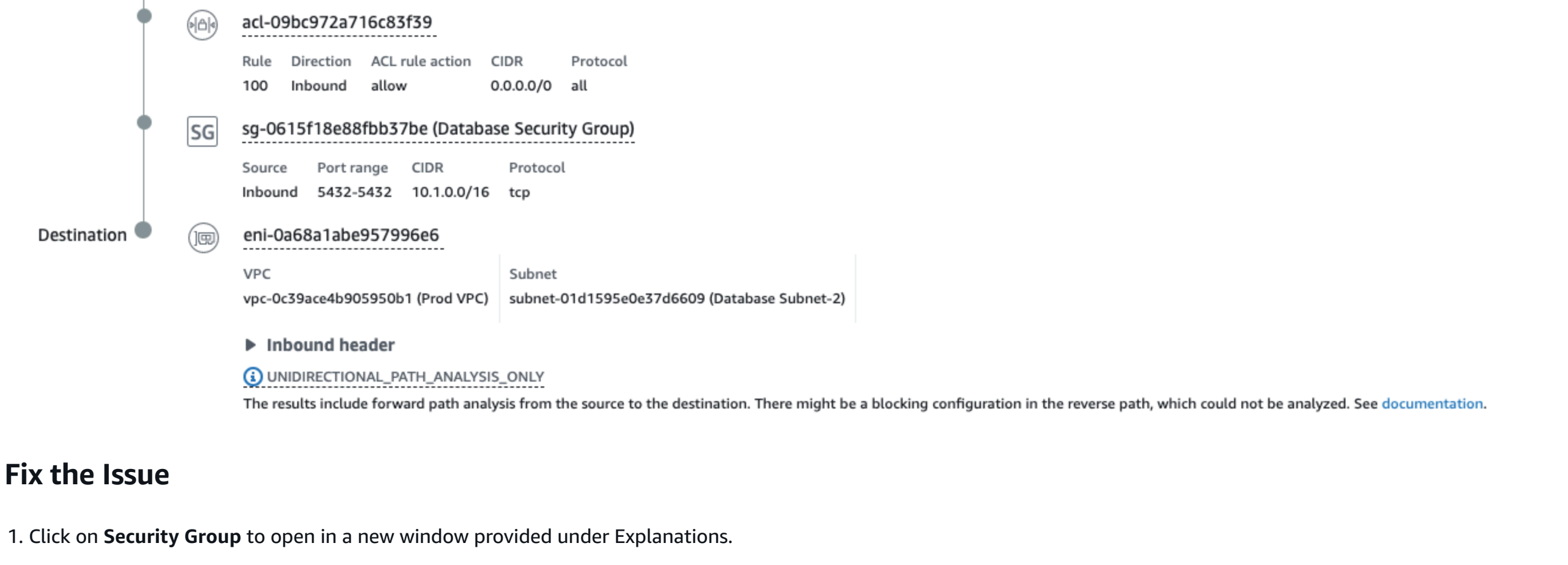


Fix the Issue

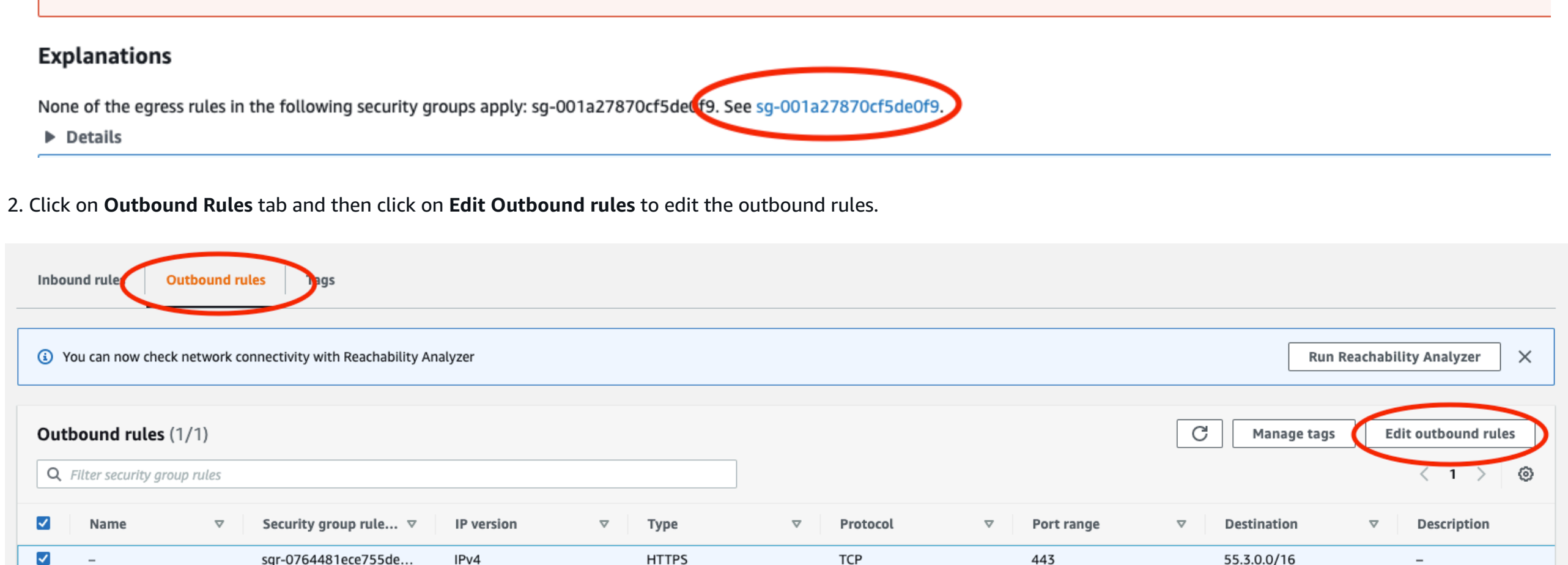
- Click on **Security Group** to open in a new window provided under **Explanations**.



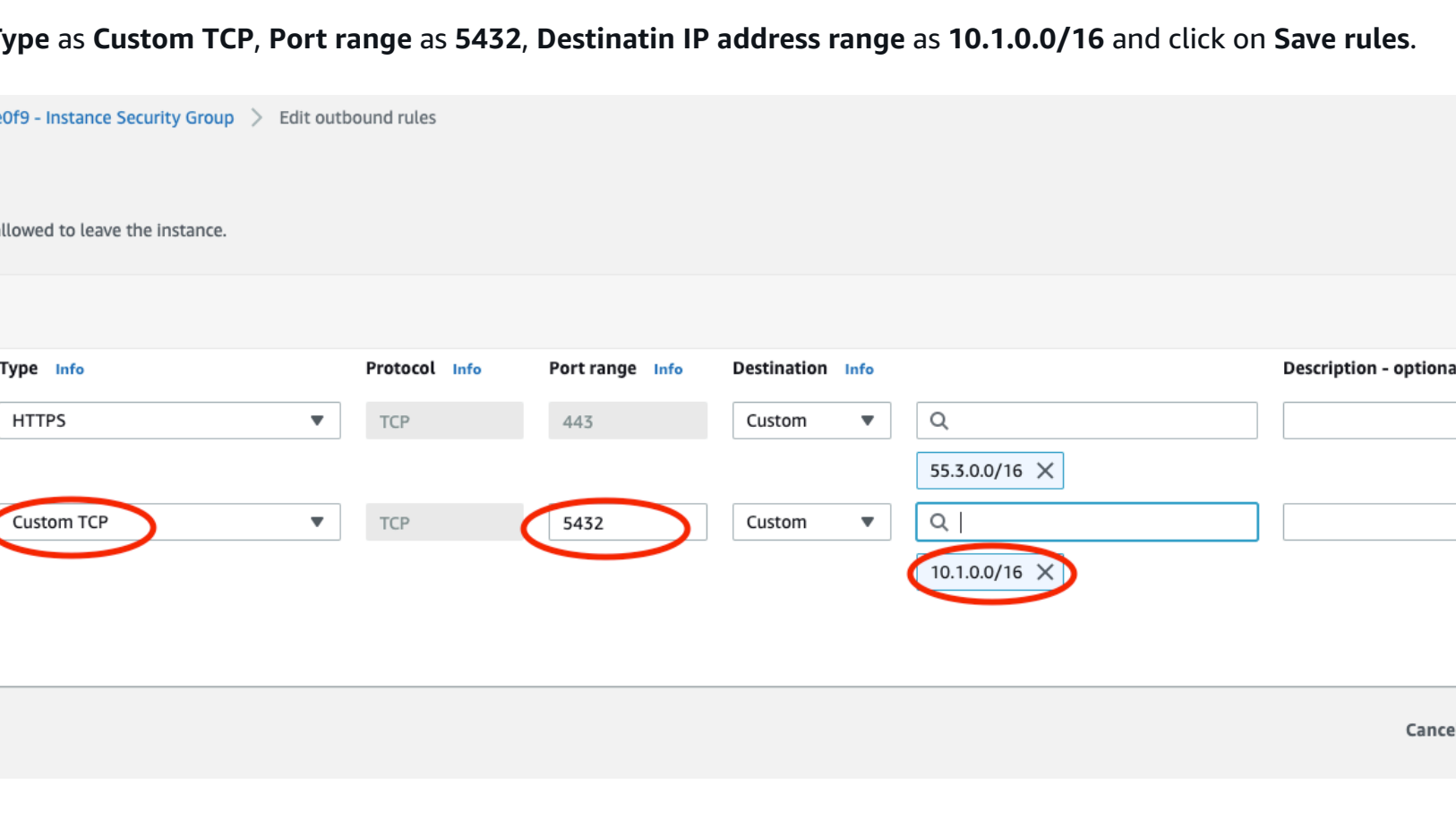
- Click on **Outbound Rules** tab and then click on **Edit Outbound rules** to edit the outbound rules.



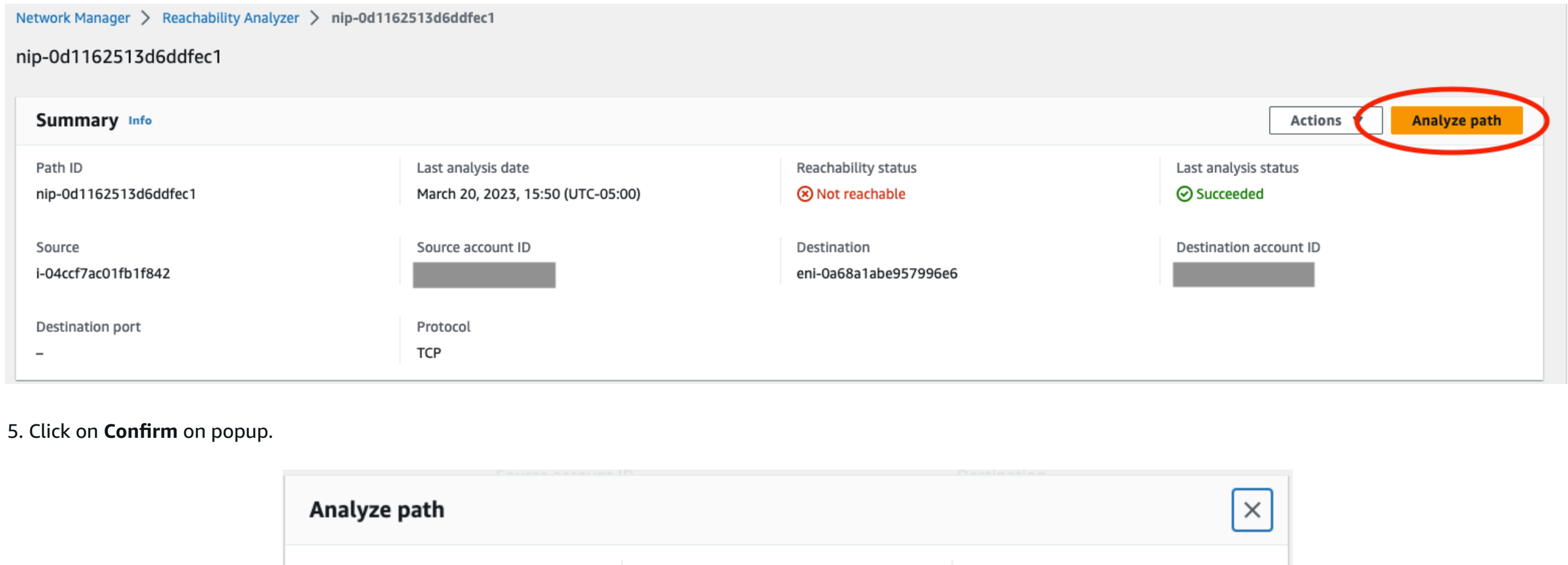
- Click on **Analyze path** to run the analysis again.



- Click on **Confirm** on popup.



- Refresh the screen and Scroll down to explore the findings once complete. **Reachability status** is **Reachable**.



Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose "Customize" or "Decline" to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose "Accept" or "Decline." To make more detailed choices, choose "Customize."

Accept

Decline

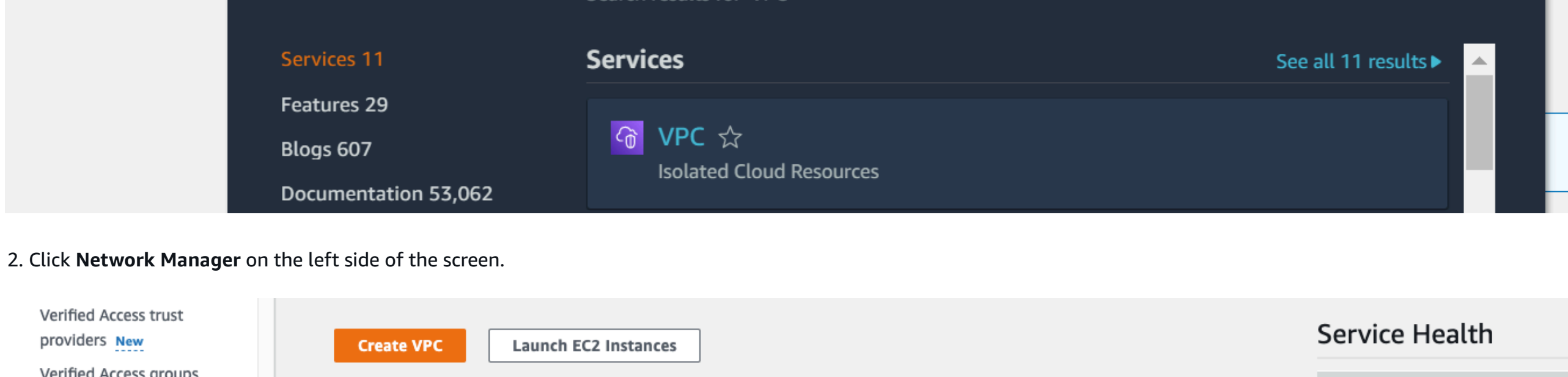
Customize

Lab 2 - Troubleshoot the connectivity issue between Prod EC2 instance in Prod VPC and Public IP address

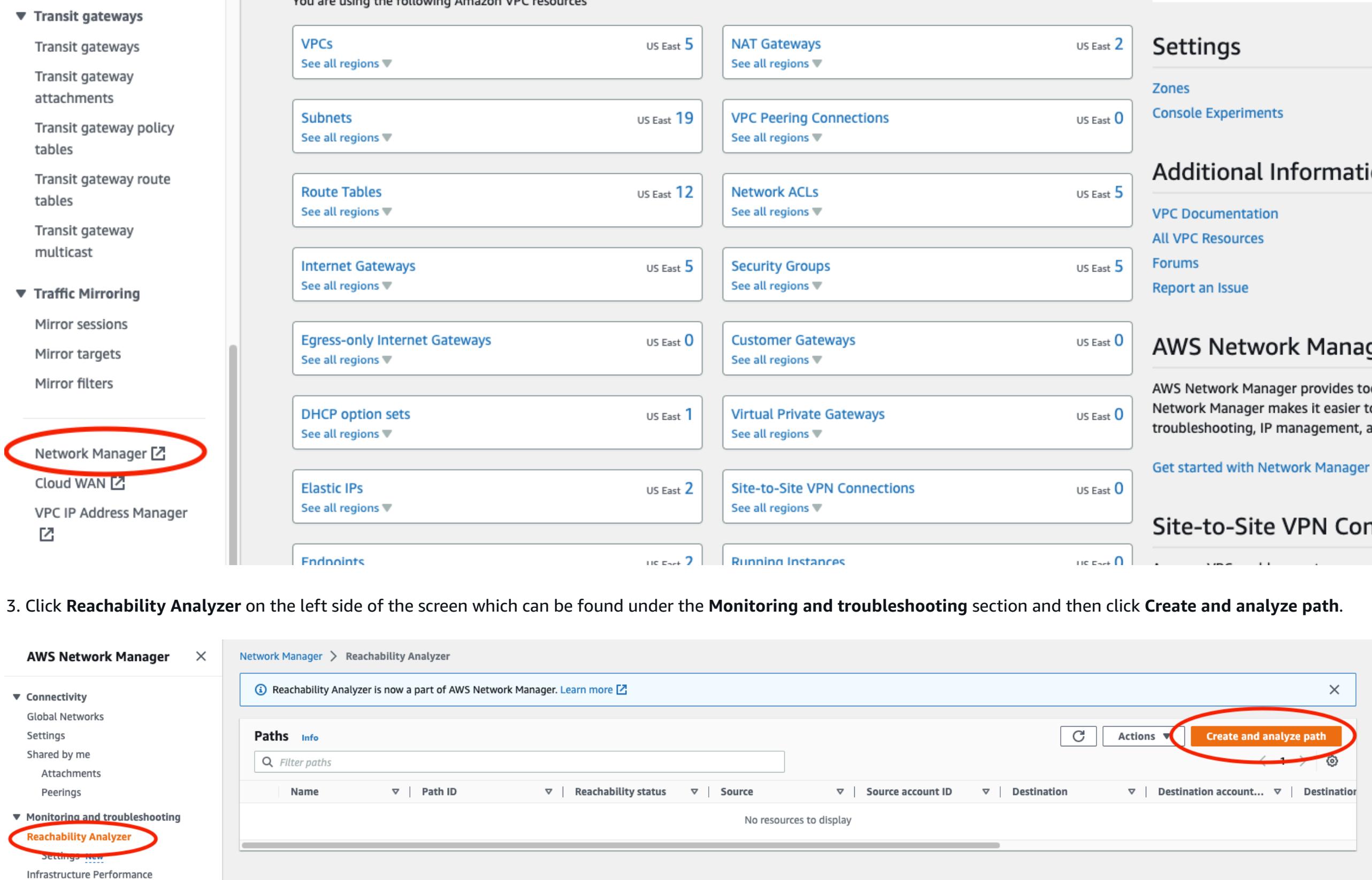
Objective: In this section of the workshop we will use Amazon VPC Reachability Analyzer to troubleshoot and fix connectivity issue between Prod EC2 instance and Public IP address which is blocked by AWS Network Firewall.

Create and Analyze path

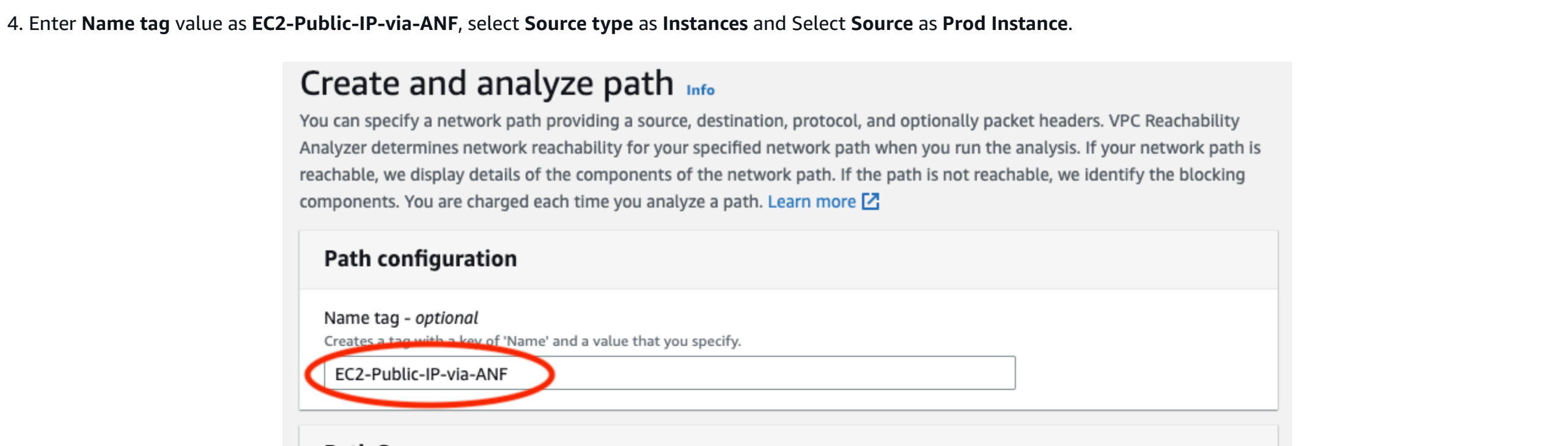
- From the **AWS Console** click in the search bar in the top left corner. Type **VPC** in the services search bar. Select **VPC** from the list.



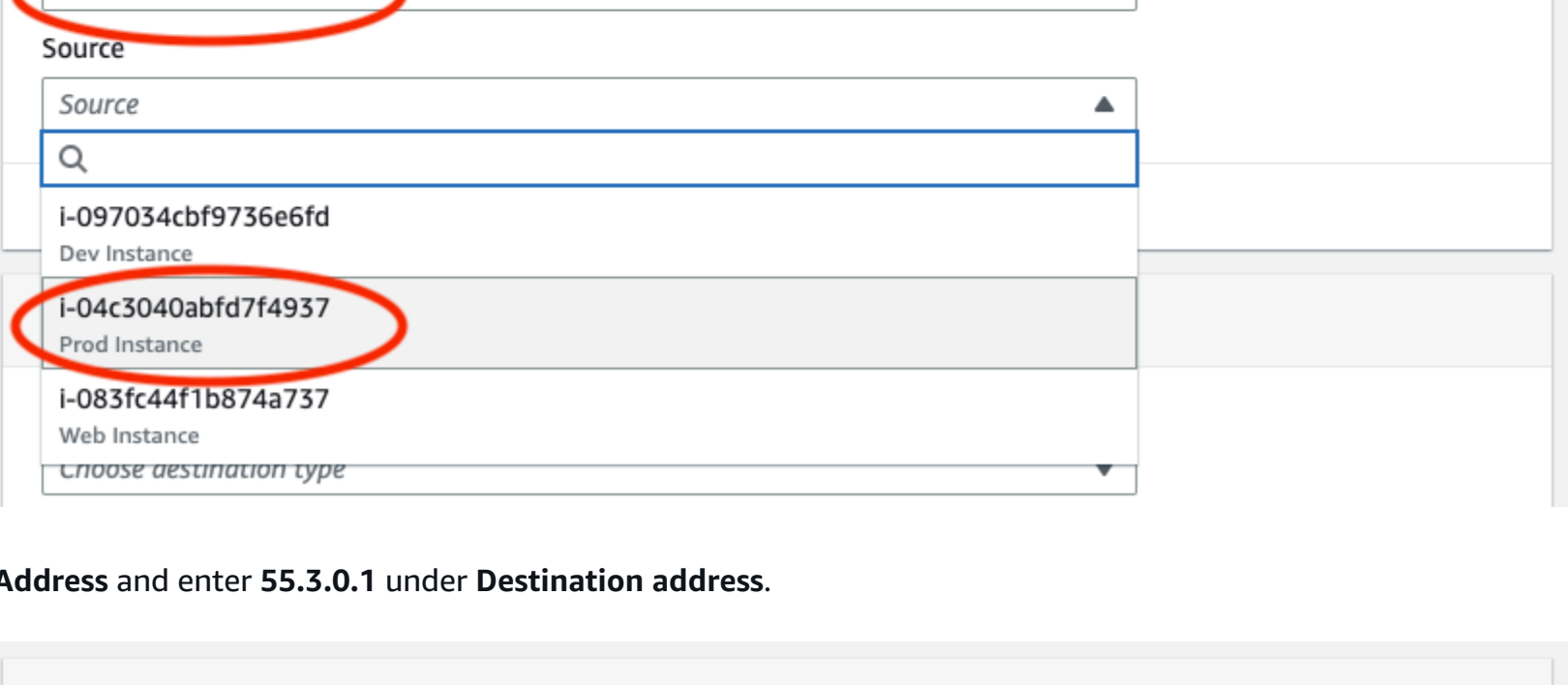
- Click **Network Manager** on the left side of the screen.



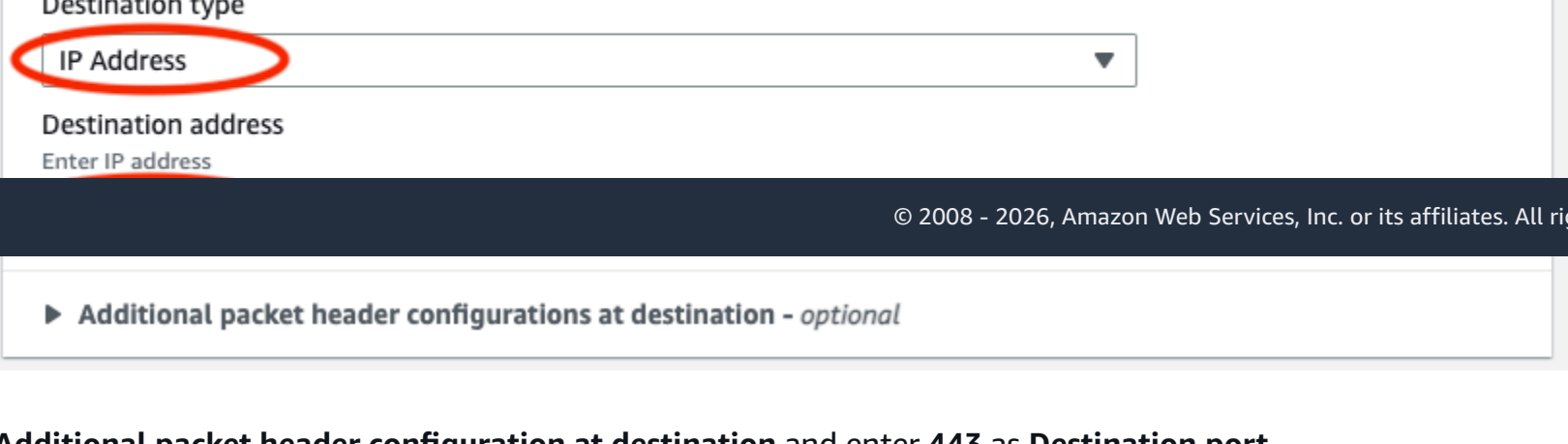
- Click **Reachability Analyzer** on the left side of the screen which can be found under the **Monitoring and troubleshooting** section and then click **Create and analyze path**.



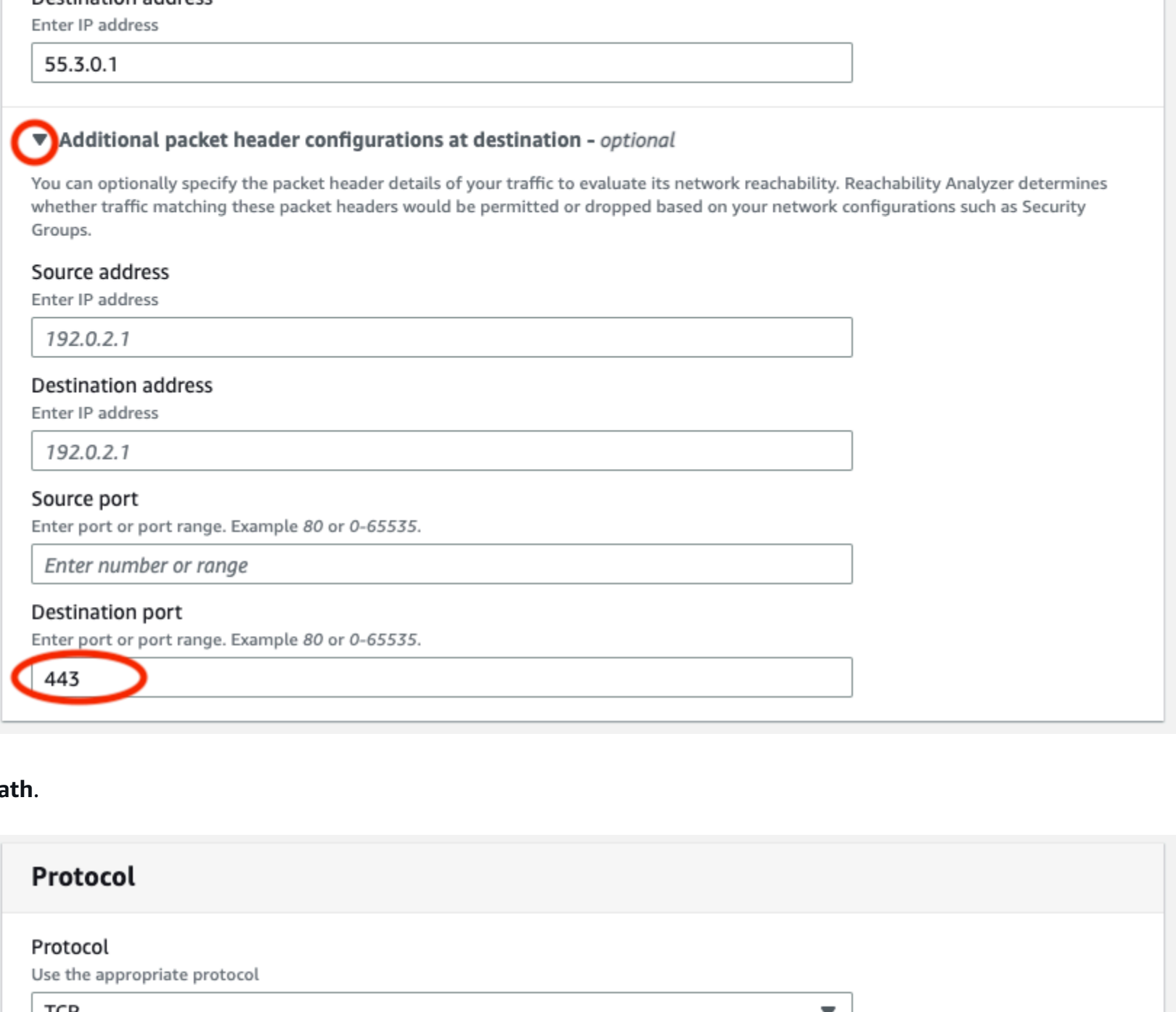
- Enter **Name tag** value as **EC2-Public-IP-via-ANF**, select **Source type** as **Instances** and Select **Source** as **Prod Instance**.



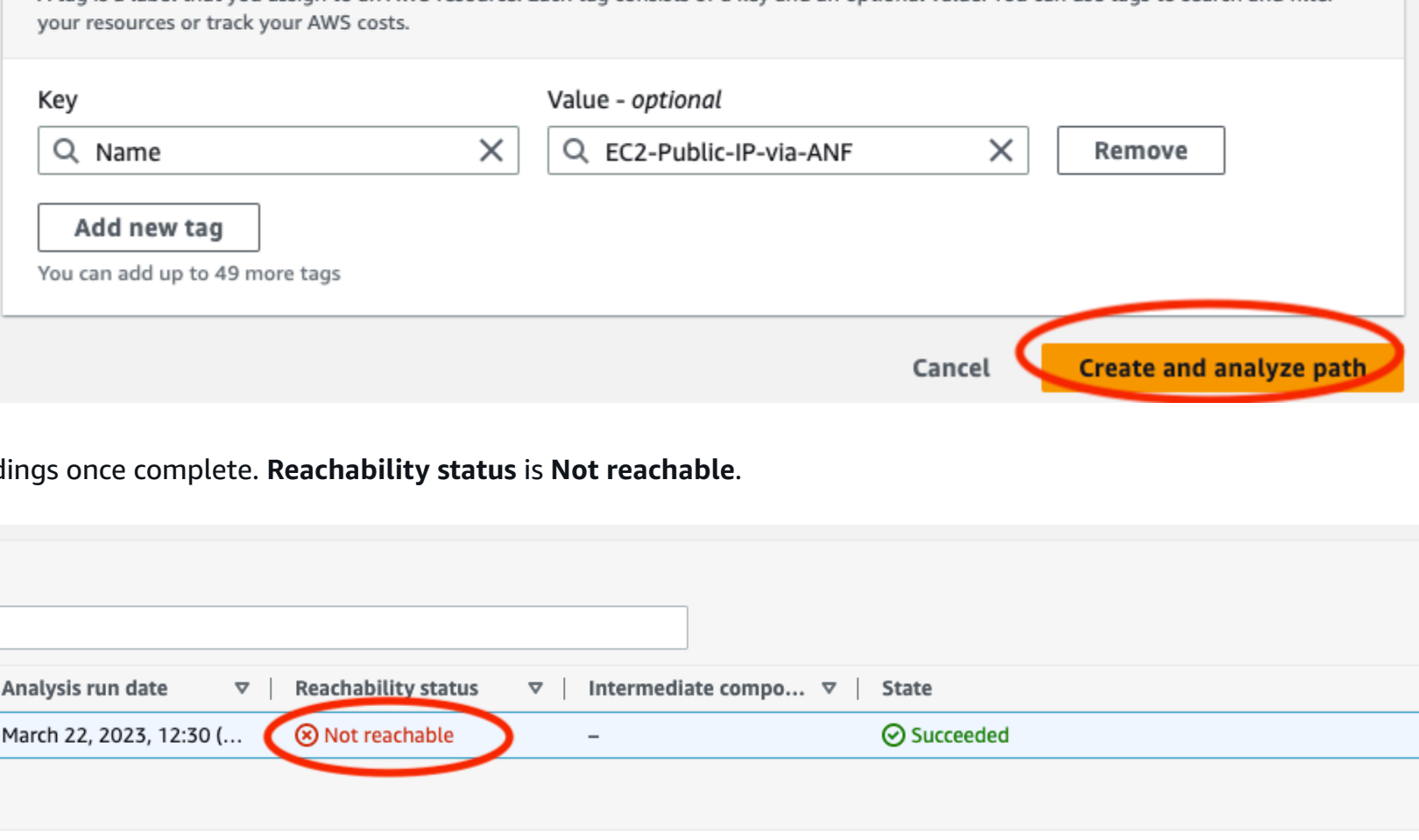
- Select **Destination type** as **IP Address** and enter **55.5.0.1** under **Destination address**.



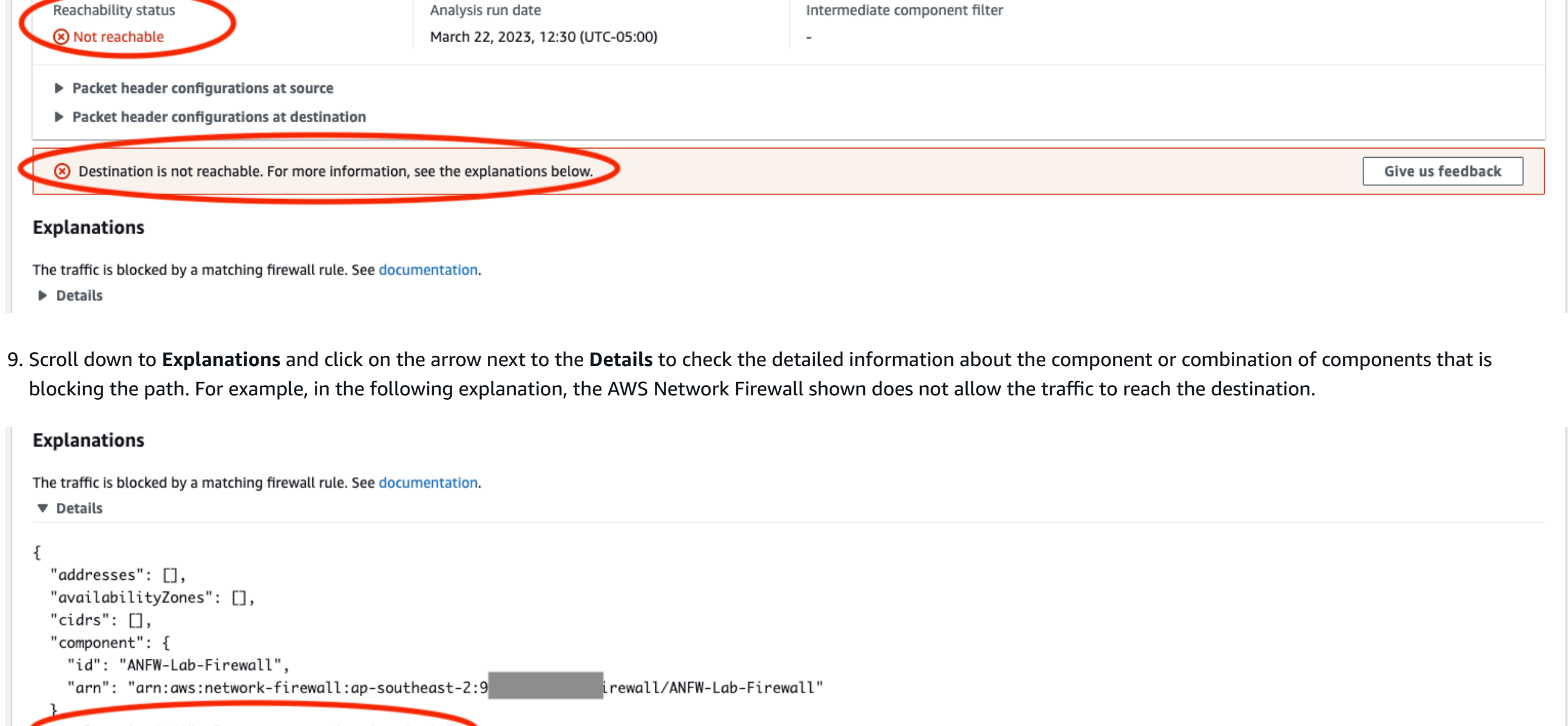
- click on the arrow next to the **Additional packet header configurations at destination** and enter **443** as **Destination port**.



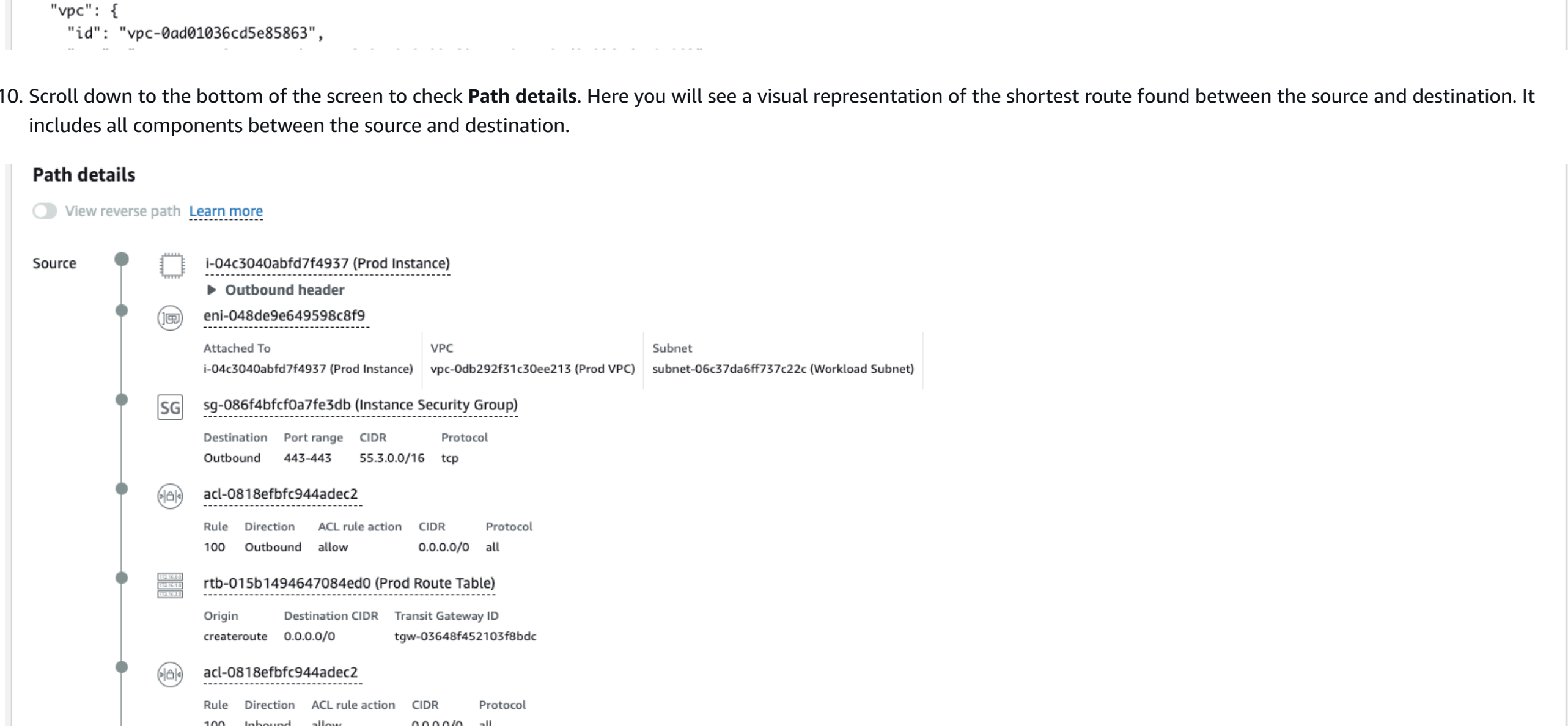
- Click on **Create and analyze path**.



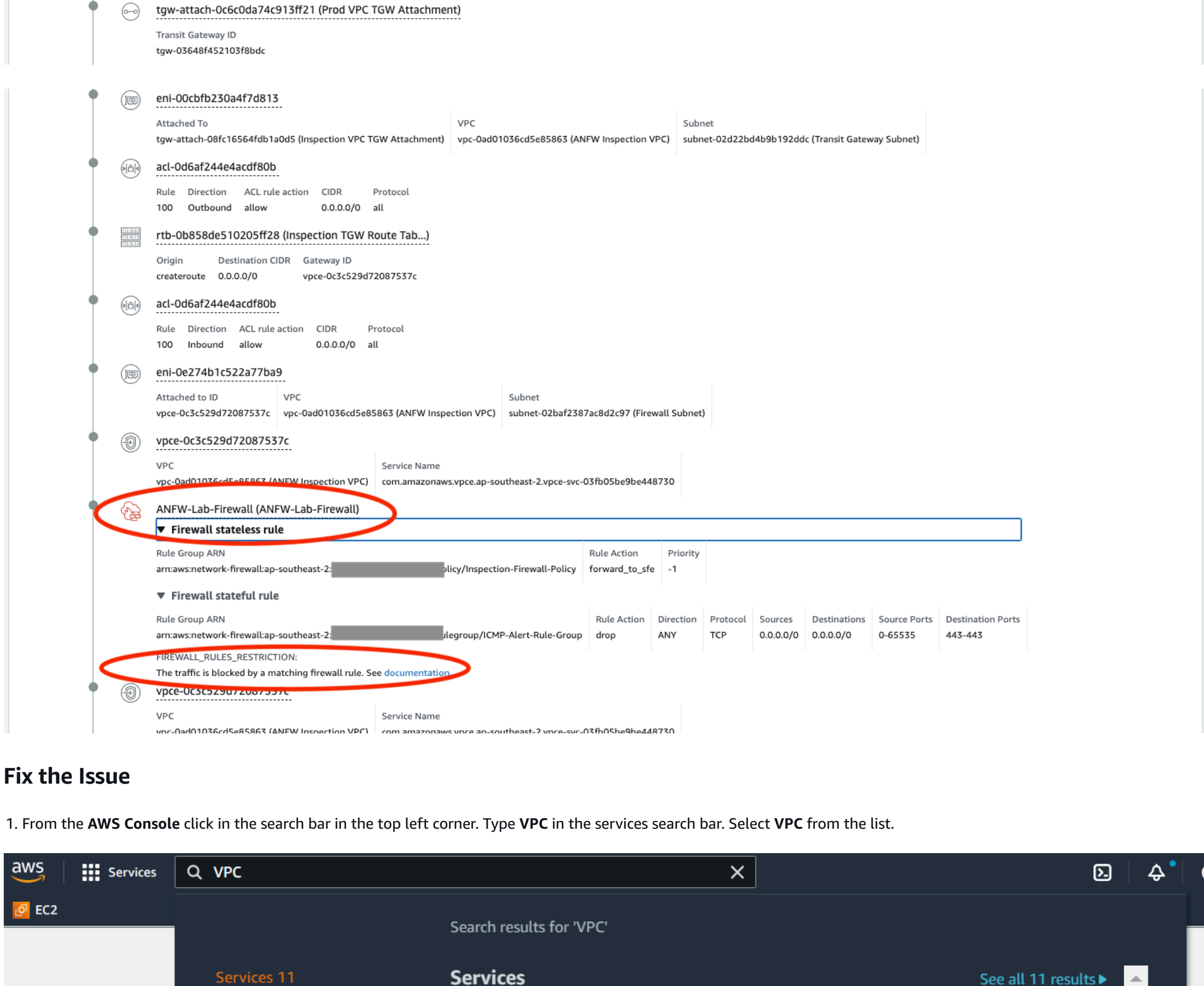
- Scroll down to explore the findings once complete. **Reachability status** is **Not reachable**.



- Scroll down to **Explanations** and click on the arrow next to the **Details** to check the detailed information about the component or combination of components that is blocking the path. For example, in the following explanation, the **AWS Network Firewall** shown does not allow the traffic to reach the destination.

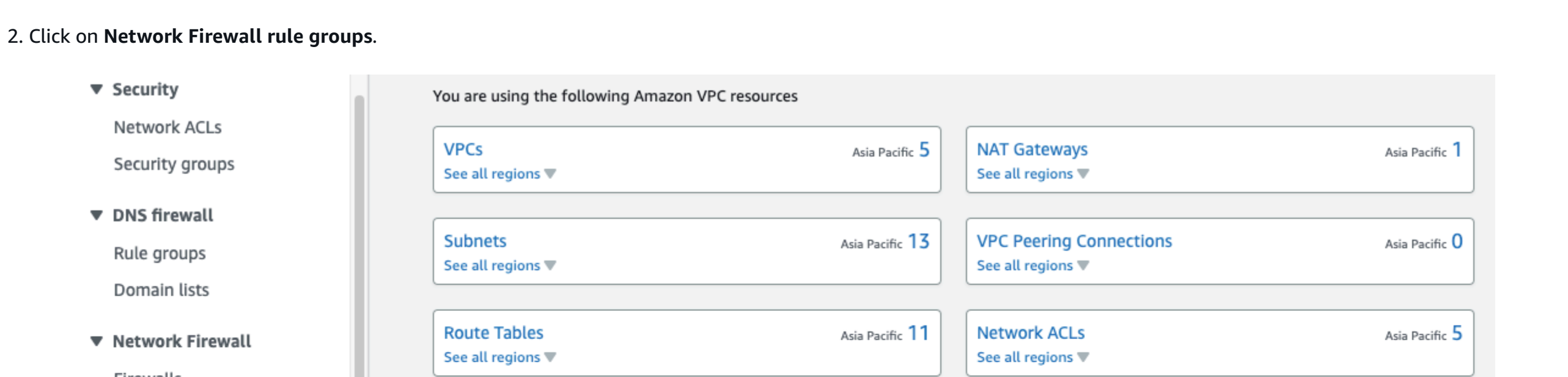


- Scroll down to the bottom of the screen to check **Path details**. Here you will see a visual representation of the shortest route found between the source and destination. It includes all components between the source and destination.

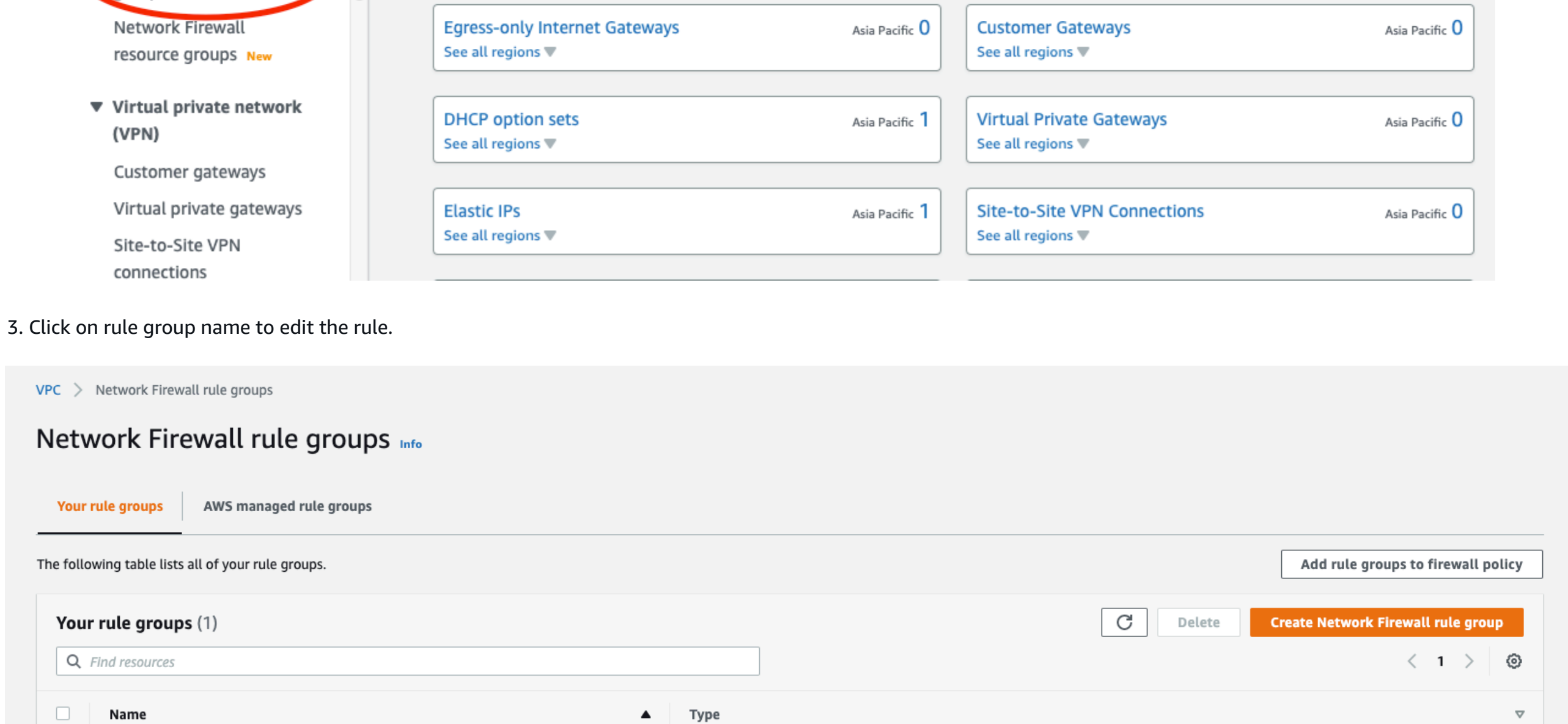


Fix the Issue

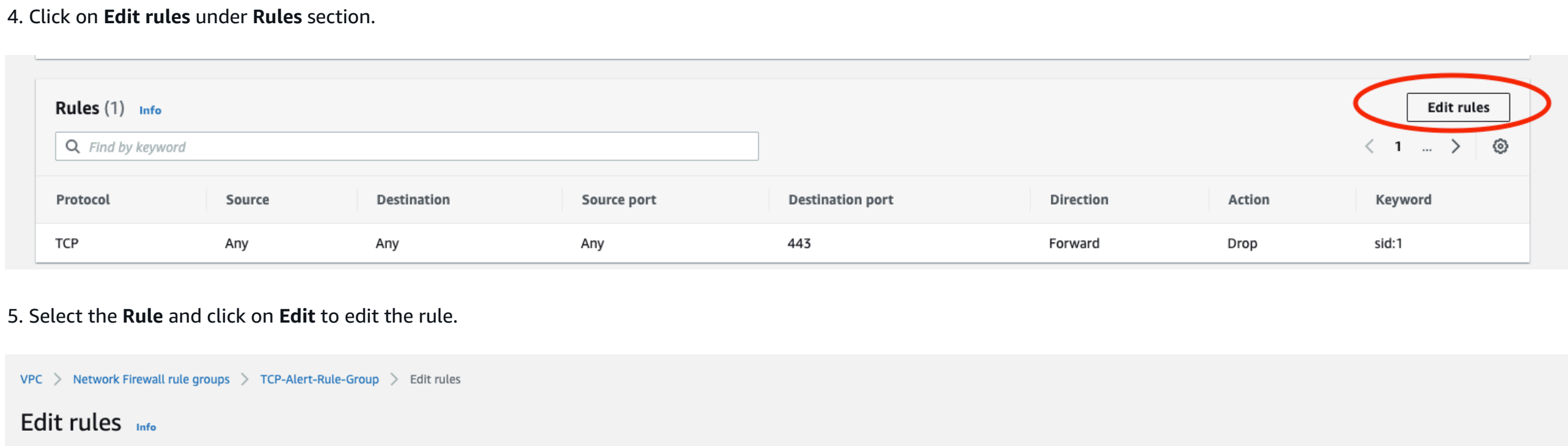
- From the **AWS Console** click in the search bar in the top left corner. Type **VPC** in the services search bar. Select **VPC** from the list.



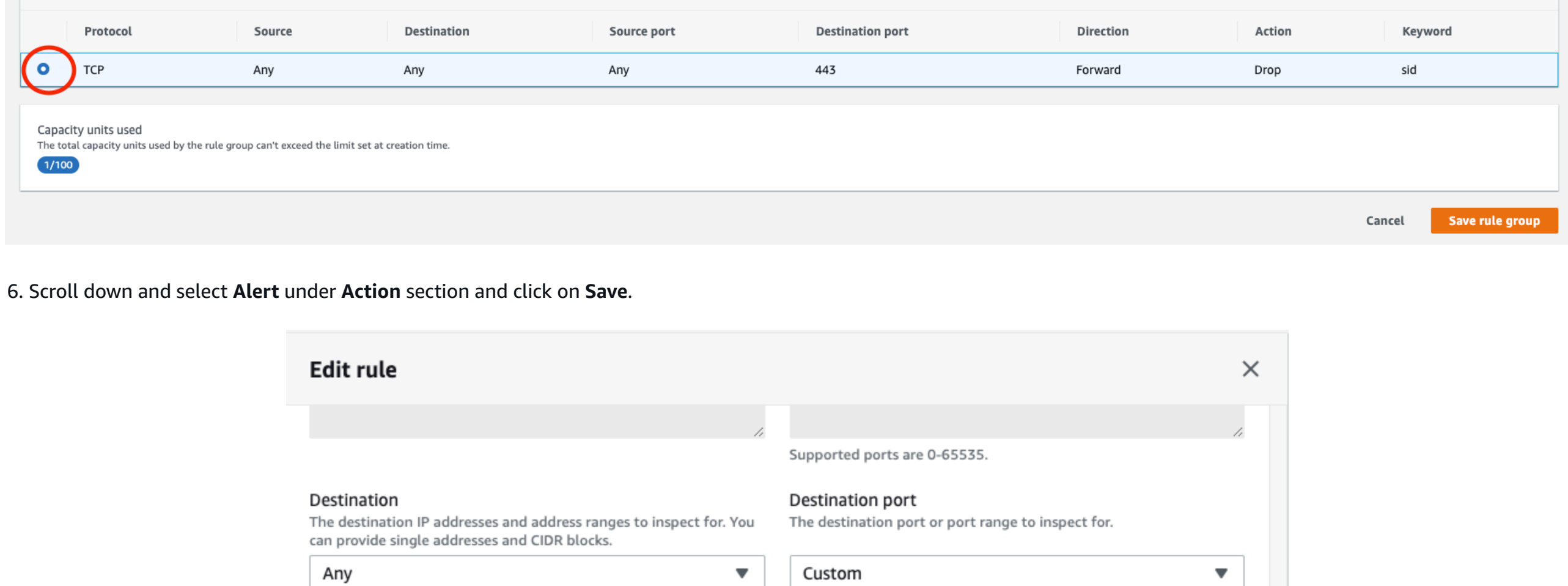
- Click on **Network Firewall rule groups**.



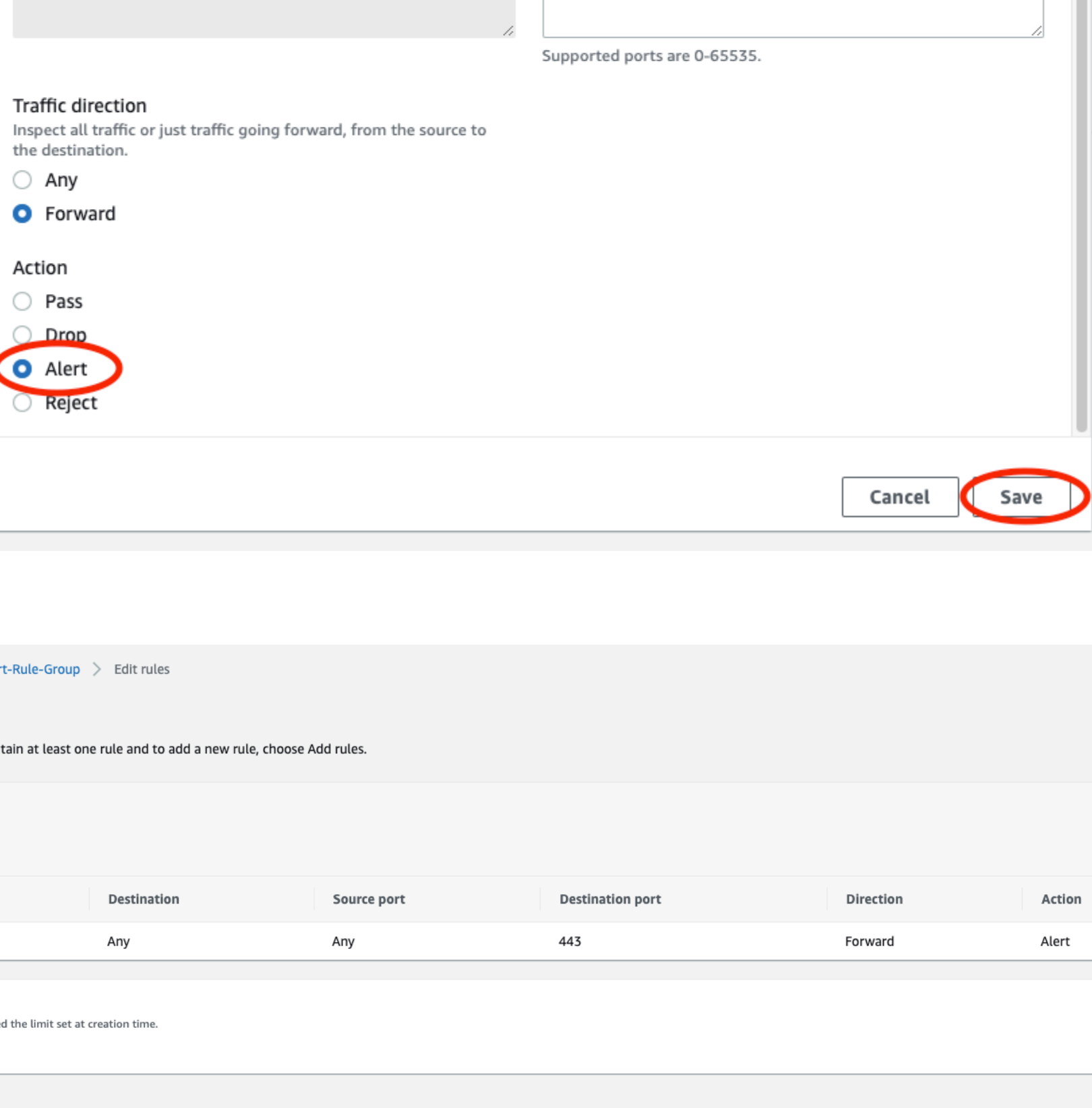
- Click on rule group name to edit the rule.



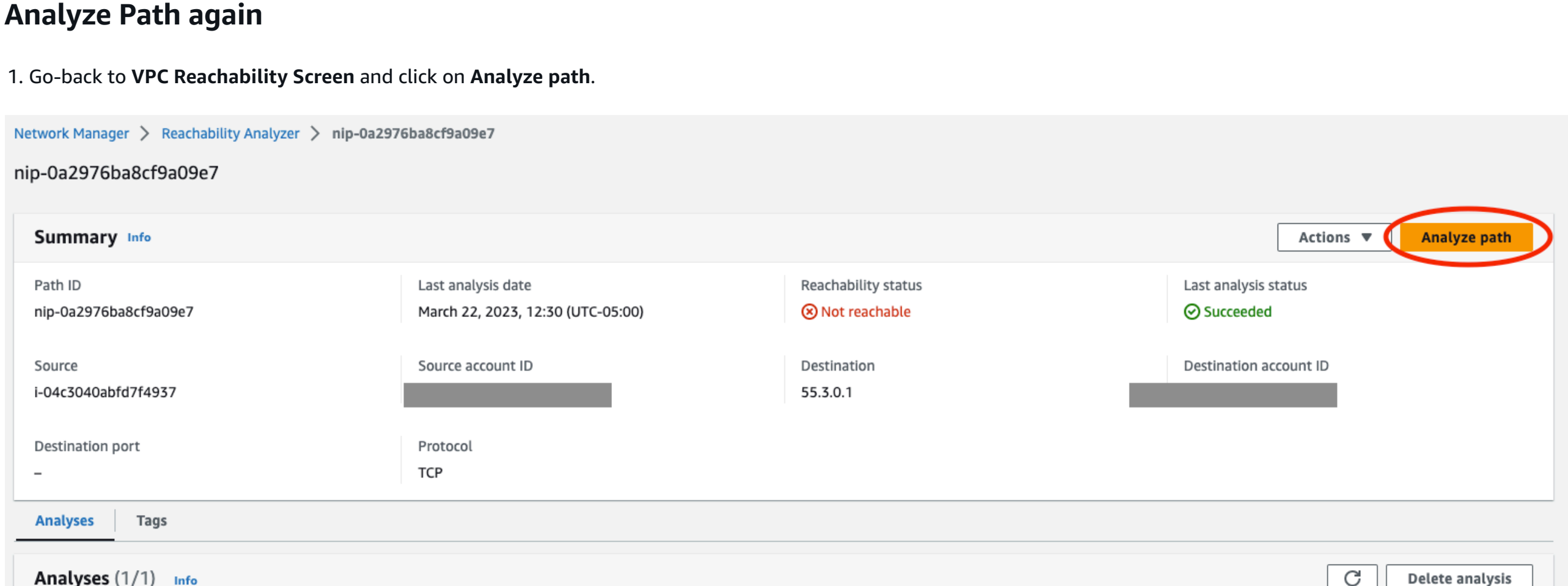
- Select the **Rule** and click on **Edit** to edit the rule.



- Scroll down and select **Alert** under **Action** section and click on **Save**.

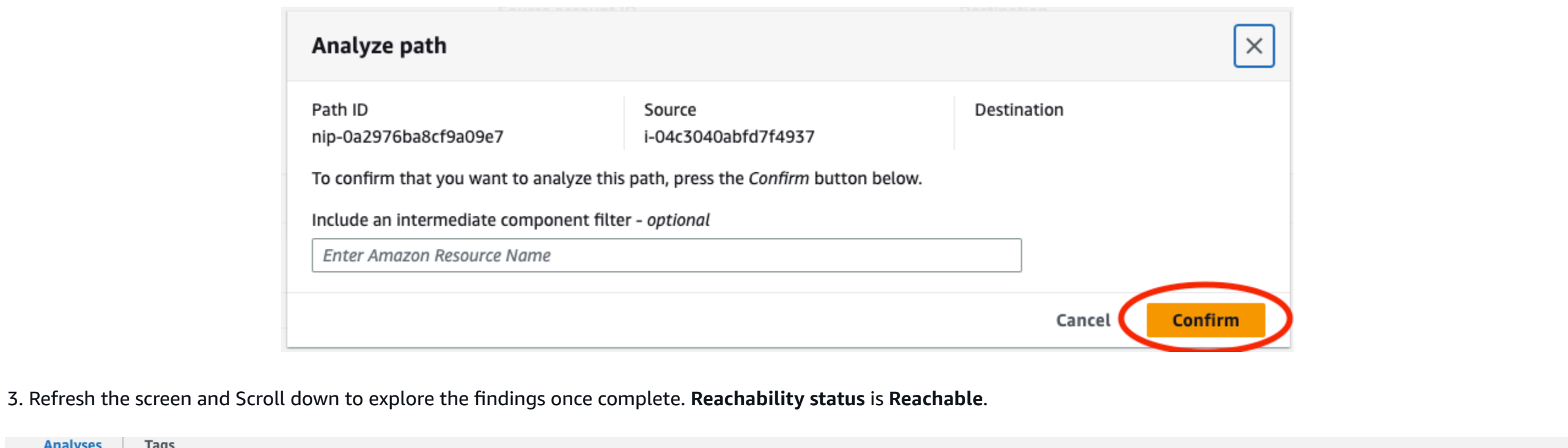


- Click on **Save rule group**.

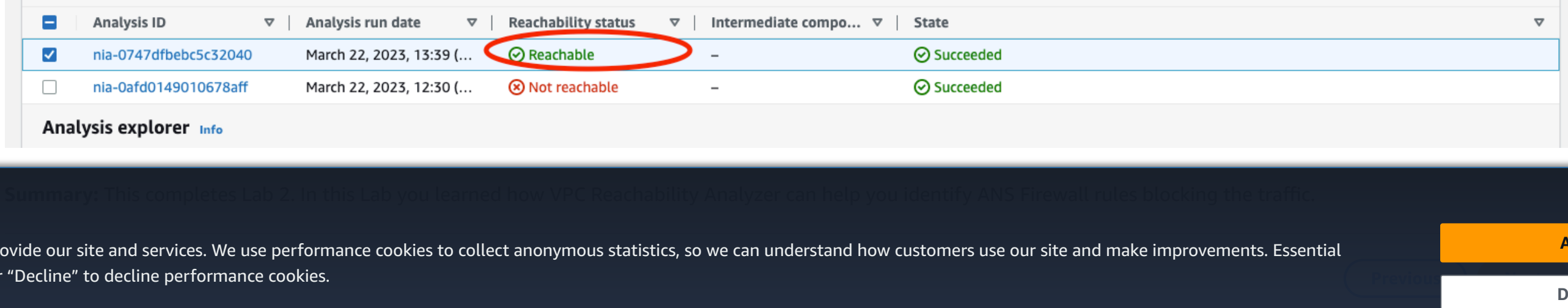


Analyze Path again

- Go-back to **VPC Reachability Screen** and click on **Analyze path**.



- Refresh the screen and Scroll down to explore the findings once complete. **Reachability status** is **Reachable**.



Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose "Customize" or "Decline" to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to define site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose "Accept" or "Decline." To make more detailed choices, choose "Customize."

Accept

Decline

Customize